React Router

💧 1️⃣ What is React Router?

👉 React Router = **Page navigation system in React**

Without it:

- React has only **one page**

With it:

- You can create multiple pages:
  - `/` → Home
  - `/about` → About
  - `/contact` → Contact

💡 Works without page reload (SPA)

⚙️ 2️⃣ Installation

Open terminal inside your React project:

```
npm install react-router-dom
```

✅ That's it — package installed

3 Simple Project Structure

```
src/
 ├─ App.js
 ├─ Home.js
 ├─ About.js
 └─ Contact.js
```

4 Create Pages

Home.js

```
function Home() {
  return <h1>Home Page</h1>;
}
export default Home;
```

About.js

```
function About() {
  return <h1>About Page</h1>;
}
export default About;
```

Contact.js

```
function Contact() {
  return <h1>Contact Page</h1>;
}
export default Contact;
```

🚀 5️⃣ Main Routing Code (App.js)

```
import { BrowserRouter, Routes, Route, Link } from "react-router-dom";
import Home from "./Home";
import About from "./About";
import Contact from "./Contact";

function App() {
  return (
    <BrowserRouter>

      {/* Navigation */}
      <nav style={{ display: "flex", gap: "20px" }}>
        <Link to="/">Home</Link>
        <Link to="/about">About</Link>
        <Link to="/contact">Contact</Link>
      </nav>

      {/* Routes */}
      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/about" element={<About />} />
        <Route path="/contact" element={<Contact />} />
      </Routes>

    </BrowserRouter>
  );
}

export default App;
```

🎯 6️⃣ How it Works (Simple)

👉 Step-by-step:

1. Click **Home / About / Contact**

2. URL changes (`/about`)
3. React finds matching `<Route>`
4. That component loads
5. Page updates **without refresh**

🧠 7️⃣ Important Concepts

Concept	Meaning
BrowserRouter	Enables routing
Routes	Group of routes
Route	One path → one component
Link	Navigation

⚡ One Line Summary

👉 **React Router = URL → Component mapping system**